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x	*y*	*Value*
		21,000
		32,000
		51,000
		64,000
		81,000
		96,000

Answer Explanation:

Apply

The task is to determine values of x and y such that $\frac{4x+8y}{x+y}=5$.

As shown, algebraic manipulation results in a simple relationship between x and y.

$$\begin{aligned}
\frac{4x+8y}{x+y} &= 5 && \text{given}\\
4x+8y &= 5(x+y) && \text{multiply both sides by } x+y\\
4x+8y &= 5x+5y && \text{apply the distributive property}\\
4x+3y &= 5x && \text{subtract } 5y \text{ from both sides}\\
3y &= x && \text{subtract } 4x \text{ from both sides}
\end{aligned}$$

***x*:**

Choose a value for x that is 3 times another value in the *Value* column. Since the value for x will be greater than the other value, it is wise to start with the greatest value, which is 96,000. Since 96,000 is 3(32,000), 96,000 is a value for x that is consistent with the given information.

The correct answer is *96,000*.

***y*:**

From the explanation above, the value of y is 32,000.

The correct answer is *32,000*.

•

Perry, Maria, and Lorna are painting rooms in a college dormitory. Working alone, Perry can paint a standard room in 3 hours, Maria can paint a standard room in 2 hours, and Lorna can paint a standard room in 2 hours 30 minutes. Perry, Maria, and Lorna have decided that, to speed up the work, 2 of them will paint a standard room together.

Select the value closest to the shortest time in which a 2-person team could paint a standard room, and select the value closest to the longest time in which a 2-person team could paint a standard room, with each person working at his or her respective rate. Make only two selections, one in each column.

Shortest time	**Longest time**	
		49 minutes
		1 hour 7 minutes
		1 hour 12 minutes
		1 hour 14 minutes
		1 hour 22 minutes
		1 hour 45 minutes

Answer Explanation:

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Apply

***Shortest time*:**

Working together, Lorna and Maria can paint a standard room in the shortest amount of time because individually they can paint a standard room in less time than Perry. Lorna can paint a standard room in $2\frac{1}{2} = \frac{5}{2}$ hours, so she can paint $\frac{2}{5}$ room in 1 hour. Maria can paint a standard room in 2 hours, so she can paint $\frac{1}{2}$ room in 1 hour. Working together, they can paint $\frac{2}{5} + \frac{1}{2} = \frac{9}{10}$ room in 1 hour. It follows that they can paint 1 room in $\frac{10}{9}$ hours, which is equivalent to $1\frac{1}{9}$ hours or about 1 hour 7 minutes.

The correct answer is *1 hour 7 minutes*.

***Longest time*:**

Working together, Lorna and Perry take the longest amount of time to paint a standard room because individually they take more time than Maria to paint a standard room. Lorna can paint a standard room in $2\frac{1}{2} = \frac{5}{2}$ hours, so she can paint $\frac{2}{5}$ room in 1 hour. Perry can paint a standard room in 3 hours, so he can paint $\frac{1}{3}$ room in 1 hour. Working together, they can paint $\frac{2}{5} + \frac{1}{3} = \frac{11}{15}$ room in 1 hour. It follows that they can paint 1 room in $\frac{15}{11}$ hours, which is equivalent to $1\frac{4}{11}$ hours or about 1 hour 22 minutes.

The correct answer is *1 hour 22 minutes*.

•

A dance show will include exactly five pieces, and each piece will be presented only once: Requiem, Smooth Step, Trampoline, Unleashed, and Waltzy. Requiem will be presented immediately before Smooth Step. Unleashed will be presented earlier than Trampoline, with at least one piece in between. Waltzy will not be the final piece presented.

In the table, select a piece that cannot be the second piece presented and a piece that cannot be the third piece presented. Make only two selections, one in each column.

Cannot be presented second	Cannot be presented third	
		Requiem
		Smooth Step
		Trampoline
		Unleashed
		Waltzy

Answer Explanation:

Infer

***Cannot be presented second*:**

One of the conditions states that Unleashed will be presented earlier than Trampoline, with at least one piece in between. It follows from this that the earliest Trampoline can be presented is third, because both Unleashed and at least one of the other pieces must be presented earlier than Trampoline. Therefore, Trampoline cannot be presented first or second.

The correct answer is *Trampoline*.

***Cannot be presented third*:**

Waltzy cannot be presented third. To see this, remember that Requiem must be presented immediately before Smooth Step. If Waltzy were third, then Requiem and Smooth Step would have to be first and second, in that order, or fourth and fifth, in that order. In either of those cases, Unleashed and Trampoline would have to be presented one right after

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and fifth, in that order. In either of those cases, Unleashed and Trampoline would have to be presented one right after the other—either first and second or fourth and fifth—but that is not allowed; there must be some piece presented between them. Therefore, Waltzy cannot be presented third.

The correct answer is *Waltzy*.

•

An American museum of East Asian art reviewed its acquisitions strategy in 2009. As part of that review, a marketing executive recommended purchasing more ceramics, jades, and paintings of people or landscapes because visitor surveys indicated that those types are of greatest interest to museumgoers in the region. The museum's curator recommended purchasing more twenty-first-century East Asian art—particularly abstract (nonrepresentational) paintings and sculptures—because the museum's collection is weakest in that area.

Among the East Asian works of art described below, select a painting and a sculpture, each of which satisfies at least one of the recommendations made by the marketing executive and at least one of the recommendations made by the curator. Make only two selections, one in each column.

Painting	Sculpture	
		Painting of girl's face, in shades of blue and red, acrylic and oil on canvas, 2006
		Chinese brush painting of fish in a tank, ink on paper, 2002
		Fish-shaped vase with dragon head, jade, 1849
		Abstract sequence of interconnected curves, ceramic, 2008
		Fossil-like, jagged sculpture suspended from cables, stainless steel, aluminum, fiberglass, and cable, 2004

Answer Explanation:

Apply

***Painting*:**

"*Painting of girl's face …*" satisfies the people or landscapes recommendation of the marketing executive and satisfies the twenty-first-century (from 2006) recommendation of the museum's curator.

The correct answer is *Painting of girl's face, in shades of blue and red, acrylic and oil on canvas, 2006*.

***Sculpture*:**

"*Abstract sequence …*" satisfies the ceramic recommendation of the marketing executive and satisfies the twenty-first-century (from 2008) recommendation of the museum's curator.

The correct answer is *Abstract sequence of interconnected curves, ceramic, 2008*.

•

Professor A: The aid industry should begin to focus its efforts to spending on primary schools in the poorest areas, to providing medicines and other basic supplies for health such as mosquito nets, and to a few key agricultural initiatives.

Professor B: Much education work has been ineffective. A village or town with poor schooling may be better off getting a road than a teacher. Once local farmers can transport produce to market, they will be willing to pay for schools—and to make sure the schools succeed.

Suppose that the professors' statements express their genuine opinions. Select statements (1) and (2) as follows: Professor A would likely disagree with (1), and Professor B would take (2) to present logical support for (1). Select only two statements, one per column.

(1)	(2)	
		The aid industry should focus less on the areas of health and agriculture than it now does.
		The aid industry should focus more on primary education than it now does.
		The aid industry should focus its spending less on primary education than it now does.
		Projects in health and agriculture are more likely to be successful if they are not paid for by the aid industry.
		Projects in education are more likely to be successful if they are paid for by the aid industry.

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		Projects in education are more likely to be successful if they are paid for by local people.

Answer Explanation:

Infer

The wording of the question suggests that the correctness of an answer for one of the response opportunities will depend on the correctness of the answer to the other. We are to find a statement in the table such that the information indicates Professor A would likely disagree with the first statement and Professor B would take the second statement to present evidence against the first.

***1*:**

The only statement in the table that the dialogue strongly suggests Professor A would disagree with is *The aid industry should focus its spending less on primary education than it now does*. For example, if the relative distribution of spending by the *aid industry* between *primary schools in the poorest areas, medicines, and other basic supplies for health such as mosquito nets*, and the *few key agricultural initiatives* is maintained, while spending in other areas is drastically reduced, then the aid industry will thereby focus its spending more on primary education than it now does.

The correct answer is *The aid industry should focus its spending less on primary education than it now does*.

***2*:**

Given the answer for *1*, the question requires identification of the statement within the table that Professor B would take to be evidence for the statement that the aid industry should focus its spending less on primary education than it now does. The claims attributed to Professor B, in the context of the dialogue with Professor A, strongly suggest that she would take such evidence to be provided by *Projects in education are more likely to be successful if they are paid for by local people*.

The correct answer is *Projects in education are more likely to be successful if they are paid for by local people*.

•

A car is traveling on a straight stretch of roadway, and the speed of the car is increasing at a constant rate with respect to time. At time 0 seconds, the speed of the car is v_0 meters per second; 10 seconds later, the front bumper of the car has traveled 125 meters, and the speed of the car is v_{10} meters per second.

Select values of v_0 and v_{10} that are together consistent with the information provided. Make only two selections, one in each column.

v_0	v_{10}	
		5
		18
		20
		36
		72

Answer Explanation:

Infer

Since the speed of the car increases at a constant rate, the car's average speed, in meters per second, over the 10-second period is equal to $\frac{1}{2}(v_0 + v_{10})$. Thus, the distance the car traveled, in meters, over that period is equal to $\frac{1}{2}(v_0 + v_{10})(10)$, or $5(v_0 + v_{10})$. But this value is given to be 125, so it must be true that $v_0 + v_{10} = \frac{125}{5}$, or 25. Because speed must be positive, both v_0 and v_{10} must be less than 25, ruling out 36 and 72 as possible values.

v_0**:**

Among the available alternatives—5, 18, and 20—the only pair that sum to 25 are 5 and 20. Since the speed is increasing, it must also be true that $v_0 \leq v_{10}$. Therefore, $v_0 = 5$.

The correct answer is *5*.

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v_{10}:

Following the analysis for Response 1, the value of v_{10} must correspond to $v_0 = 5$. So, $5 + v_{10} = 25$, and therefore, $v_{10} = 20$.

The correct answer is *20*.

•

For each positive integer n, the quantity s_n is defined such that $s_{n+2} = (s_n)^2 - s_{n+1}$. In addition, $s_2 = 1$.

Select values for s_1 and s_4 that are jointly compatible with these conditions. Select only two values, one in each column.

s_1	s_4	
		−12
		−7
		−3
		−1
		0

Answer Explanation:

Infer

For $n = 1$, the equation $s_{n+2} = (s_n)^2 - s_{n+1}$ is equivalent to $s_3 = (s_1)^2 - s_2$. For $n = 2$, the equation $s_{n+2} = (s_n)^2 - s_{n+1}$ is equivalent to $s_4 = (s2)^2 - s_3$, and substituting $(s_1)^2 - s_2$ for s_3 yields $s_4 = (s_2)^2 - (s_1)^2 + s_2$. (Continuing in this fashion, it is possible to produce, for each positive integer n, an expression for s_n in terms of s_1 and s_2.) Substituting 1 for s_2 in the equation for s_4, it follows that the values of s_1 and s_4 are related by the equation $s_4 = 2 - (s_1)^2$. One strategy is to substitute each value in the table for s_1 to see if the corresponding value of s_4 is present.

s_1:

Because the value of $(s_1)^2$ is nonnegative and increases rapidly as the absolute value of s_1 increases, it makes sense to begin testing possible values for s_1 at the bottom of the table. When s_1 is equal to 0 or −1, the quantity $2 - (s_1)^2$ is positive, so the corresponding value of s_4 does not appear. When $s_1 = -3$, the corresponding value of s_4 is $2 - (-3)^2 = -7$, which is present in the table. Note that when $s_1 \leq -7$, it follows that $2 - (s_1)^2 \leq 2 - 49$, so the corresponding value of s_4 is less than or equal to −47 and hence cannot appear in the table. So the value of s_1 must be −3.

The correct answer is −3.

s_4:

Following the analysis for Response 1, the value of s_4 must correspond to $s_1 = -3$, and it is given by $s_4 = -7$.

The correct answer is −7.

•

The *finesse* of a gold bar is the weight of the gold present in the bar divided by the total weight of the bar. A jeweler has three gold bars. The three bars have identical weights, and the first two bars have identical finesse. If all three bars were melted and combined into one bar, the finesse of the resulting bar would be 0.96.

Select a finesse of the *First two bars* and a finesse of the *Third bar* that are jointly consistent with the given information. Make only two selections, one in each column.

First two bars	Third bar	
		0.92
		0.93
		0.94

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		0.94
		0.98
		0.99

Answer Explanation:

Infer

***First two bars*:**

According to the text, the finesse of a gold bar is equal to the weight of gold in the bar divided by the total weight of the bar. The text also indicates that the three gold bars have the same weight, that the first two bars have the same finesse, and that melting and combining the three bars would result in a gold bar with a finesse of 0.96. Since the first two bars have identical finesse and identical weight, the weight of the gold present in those bars must also be identical. Let x equal the weight of the gold in each of the first two bars, let y equal the weight of the gold in the third bar, and let w be the total weight of each bar. Then, since the finesse of the combined bar is 0.96, it must be the case that $\frac{2x+y}{3w} = 0.96$, or $2\left(\frac{x}{w}\right) + \left(\frac{y}{w}\right) = \left(0.96\right)\left(3\right) = 2.88$. But then if $F_1 = \left(\frac{x}{w}\right)$ is the finesse of each of the first two bars, and $F_3 = \left(\frac{y}{w}\right)$ is the finesse of the third bar, then $2F_1 + F_3 = 2.88$. Among the available options, substituting 0.98 for F_1 and 0.92 for F_3 gives us 2(0.98) + 0.92 = 1.96 + 0.92 = 2.88. No other pair of options yields the correct result.

The correct answer is *0.98*.

***Third bar*:**

Based on the explanation for Response 1, the correct answer for Response 2 is 0.92.

The correct answer is *0.92*.

•

A fair coin has 2 distinct flat sides—one of which bears the image of a face and the other of which does not—and when the coin is tossed, the probability that the coin will land face up is $\frac{1}{2}$. For certain values of M, N, p, and q, when M fair coins are tossed simultaneously, the probability is p that all M coins land face up, and when N fair coins are tossed simultaneously, the probability is q that all N coins land face up. Furthermore, $M < N$ and $\frac{1}{p} + \frac{1}{q} = 72$.

In the table, select a value for M and a value for N that are jointly consistent with the given information. Make only two selections, one in each column.

M	N	
		2
		3
		4
		5
		6
		7

Answer Explanation:

Infer

If M is a positive integer, then the probability, p, that M fair coins all land face up is $p = \left(\frac{1}{2}\right)^M = \frac{1}{2^M}$

Similarly, if N is a positive integer, then the probability, q, that N fair coins all land face up is

$q = \left(\frac{1}{\ }\right)^N = \frac{1}{\ }$

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$q = \left(\frac{}{2}\right) \quad = \frac{}{2^N}$

Since $\frac{1}{p} + \frac{1}{q} = 72$

Then $p = \frac{1}{\left(\frac{1}{2^M}\right)} + \frac{1}{\left(\frac{1}{2^N}\right)} = 2^M + 2^N = 72$

The task, then, is to find two powers of 2 (an Mth power and an Nth power) that add up to 72. The correct answer for M will be the lesser of the two exponents. Note that for the given options, the only possible powers of 2 are 4, 8, 16, 32, 64, and 128. The only way for two of these to add up to 72 is for one to be $8 = 2^3$ and the other to be $64 = 2^6$. Since M is to be the lesser of the two exponents and N is to be the greater of the two exponents, it must be that $M = 3$ and $N = 6$.

***M*:**

As noted in the discussion above, $M = 3$.

The correct answer is *3*.

***N*:**

As noted in the discussion above, $N = 6$.

The correct answer is *6*.

•

Educational policy analyst: To improve the long-term economic benefits that our country's universities provide to their students, the government should provide subsidies enabling the universities to hire more academic staff. Hiring more academic staff would allow smaller class sizes. Statistically, university students in smaller classes tend to receive higher grades than those in larger classes. And on average, a student who earns higher grades in university classes tends to have a higher salary after graduation than a student who earns lower grades. Thus, hiring more academic staff nationwide should improve the economic well-being of graduates nationwide.

Consider the incomplete statement:

According to the analyst's argument, students' ___1___ is bolstered by their ___2___.

Select for *1* and *2* the options that complete the statement in the manner that most accurately reflects the given information. Make only two selections, one in each column.

1	2	
		tendency to earn higher grades in university classes
		economic well-being after graduating from university
		being in larger than average university classes
		receiving better education overall
		learning from a greater number of academic staff

Answer Explanation:

Evaluate

We are asked to complete the statement so that it is an accurate reflection of the information provided in the educational policy analyst's argument.

The educational policy analyst asserts that *hiring more academic staff would allow smaller class sizes*, and that *on average, a student who earns higher grades in university classes tends to have a higher salary after graduation than a student who earns lower grades*. This leads the analyst to conclude that *hiring more academic staff would improve the economic well-being of graduates nationwide*. Given this conclusion in conjunction with the analyst's other assertions, according to the analyst's argument, *students' economic well-being after graduation from university* is bolstered by their *tendency to earn higher grades in university classes*.

The analyst makes no claims about the quality of education overall or the number of academic staff students learn

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The analyst makes his claims about the quality of education overall or the number of academic staff students learn from. Furthermore, the only implication with respect to larger university class sizes is that they would likely not bolster either grades or salaries. Therefore, no other pair of options completes the statement so that it is an accurate reflection of the information provided.

***1*:**

The correct answer is *economic well-being after graduating from university*.

***2*:**

The correct answer is *tendency to earn higher grades in university classes*.

•

The *payload rating* (PR) of a truck is the truck's recommended load weight, which is specified as a number of tonnes (t), where 1 tonne = 1,000 kilograms. A certain truck's PR is a whole number of tonnes. The truck has hauled exactly 7 loads, exactly 3 of which had a greater weight than the truck's PR. The weights of these 7 loads, in tonnes, are as follows:

50, 51, 52, 52, 54, 54, 56.

Select the *Least* and the *Greatest* possible values for the truck's PR, in tonnes. Make only two selections, one in each column.

Least	Greatest	
		49
		50
		51
		52
		53
		54

Answer Explanation:

Apply

***Least*:**

According to the information provided, the PR is a whole number of tonnes, and among the seven loads (weighing 50, 51, 52, 52, 54, 54, and 56 tonnes), exactly three exceeded the truck's PR. Given this, the PR must be a whole number of tonnes less than 54 and no less than 52, so the truck's PR can be either 52 or 53 tonnes. Therefore, the least possible value for the truck's PR is 52 tonnes.

The correct answer is *52*.

***Greatest*:**

Based on the explanation for Response 1, the greatest possible value for the truck's PR is 53 tonnes.

The correct answer is *53*.

•

Each of the species of a certain type of insect has at least one of five significant traits: A, B, C, D, and E. Furthermore, one study has determined that any species with Trait A has at least one of the Traits B and C. Another study has determined that any species with Trait C has at least one of the Traits D and E. And a third study has determined that any of the species with Trait B has Trait C. The results of each study are correct.

Select different options for X and for Y such that the following statement most accurately describes the passage. Make only two selections, one in each column.

In addition to the relationship between Traits B and C that is stated explicitly with respect to the third study, the passage implies that any of the insects that has Trait ____X ____ *also has Trait* ____Y____.

X	Y	
		A
		B

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		B
		C
		D
		E

Answer Explanation:

Infer

We are asked to identify, in addition to the relationship between Traits B and C, a relationship between two of the five significant traits of a certain type of insect that is implied by the information in the passage.

The passage states that any species of insect with Trait A has at least one of the Traits B and C—any insects with Trait A will have Trait B, Trait C, or both. The passage also states that any of the species with Trait B has Trait C. Taken together, these claims imply that any of the insects that has Trait A also has Trait C—an insect with Trait A will either have Trait C and not Trait B, or it will have both Trait B and Trait C. The passage does not imply any other relationships between the five significant traits such that any insect that has any one of the significant traits must also have another of the significant traits. Therefore, the statement that the passage implies that any of the insects that has Trait A also has Trait C most accurately describes the passage.

***X*:**

The correct answer is *A*.

***Y*:**

The correct answer is *C*.

•

Events A, B, and C have the following probabilities of future occurrence:

A: 20%

B: 50%

C: 80%

On the basis of these probabilities, select for *Both A and B* a description that must be true of the probability that both A and B will occur. And select for *B or C or Both* a description that must be true of the probability that B or C or both will occur. Make only two selections, one in each column.

Both A and B	B or C or Both	
		Must be equal to 40%
		Must be equal to 50%
		Must be equal to 70%
		Must be less than or equal to 20%
		Must be greater than or equal to 80%

Answer Explanation:

Apply

***Both A and B*:**

The probability of two events both occurring never exceeds either individual event's probability of occurring. The probability of A occurring is only 20%, so the probability of both A and B occurring cannot exceed 20%.

The correct answer is *Must be less than or equal to 20%*.

***B or C or Both*:**

The probability that at least one of two events will occur is never less than either individual event's probability of occurring. The probability of C occurring is 80%, so the probability that either B or C or both will occur must be at least 80%.

The correct answer is *Must be greater than or equal to 80%*.

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•

Premiolex Corporation spokesperson: In our survey of Premiolex customers, the majority of respondents rated our services as “Excellent,” whereas in our competitor Cretazole’s survey, the majority of their respondents rated Cretazole’s services as “Very good.” That’s proof that—on average, at least—our customers have a higher opinion of Premiolex’s services than Cretazole’s customers have of its services.

Skeptic: I don’t think the spokesperson’s conclusion necessarily follows from the evidence he cites. What if, for example, the Premiolex survey asked participants to choose from among options that weren’t all identical to the ones used in Cretazole’s survey?

Select for *Premiolex survey* and *Cretazole survey* the options such that, in combination, the Premiolex survey and Cretazole survey would support the skeptic’s position and be consistent with the spokesperson’s report of the results but indicate that the spokesperson’s conclusion does not necessarily follow. Make only two selections, one in each column.

Premiolex Survey	**Cretazole Survey**	**Condition: 1,000 respondents were asked to choose among these survey options**	**Results: Number of respondents who chose each survey option**
		Excellent	517
		Average	482
		Below average	1
		Poor	0
		Unacceptable	0
		Excellent	463
		Average	501
		Below average	0
		Poor	35
		Unacceptable	1
		Excellent	454
		Average	511
		Below average	34
		Poor	0
		Unacceptable	1
		Excellent	512
		Average	478
		Below average	10
		Poor	0
		Unacceptable	0
		Excellent	23
		Average	516
		Below average	230
		Poor	231
		Unacceptable	0

Answer Explanation:

Evaluate

Only two selections are consistent with the Premiolex Corporation spokesperson’s report that more than half of the respondents to the Premiolex Survey rated Premiolex’s services as “Excellent”—the first selection (517 of 1,000 ratings were “Excellent”) and the fourth selection (512 of 1,000 ratings were “Excellent”). Also, only slightly more than half for the first selection rated Premiolex’s services above average (because the only survey option for above

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average was "Excellent"), whereas nearly all for the fourth selection rated Premiolex's services above average (512 + 478 = 990 ratings were for above average). Therefore, among these two selections, the choice of the **first selection** for the Premiolex Survey options offers the greater possibility of failure for the spokesperson's conclusion.

Neither the first selection nor the second selection could be the survey options for the Cretazole Survey because neither selection has "Very good" as a survey option. Also, the fourth selection could not be the survey options in the Cretazole Survey because, for this selection, a majority of respondents did not choose "Very good" (478 of 1,000 ratings were "Very good"). Of the two remaining selections for the Cretazole Survey options, the **third selection** offers a greater possibility of failure than the fifth selection, because the fifth selection has a relatively large number of respondents who chose "Poor."

Using the first selection for the Premiolex Survey, the majority of respondents chose "Excellent," and using the third selection for the Cretazole Survey, the majority of respondents chose "Very good." Thus, these two selections are consistent with the spokesperson's report of the results. However, for the first selection, only 517 respondents chose a rating higher than average, whereas for the third selection, 454 + 511 = 965 respondents chose a rating higher than average. Therefore, it is possible that, on average, the Premiolex respondents did NOT have a higher opinion of Premiolex's services than Cretazole's customers had of its services. That is, the spokesperson's conclusion does not necessarily follow.

Premiolex Survey:

The correct answer is *first selection*.

Cretazole Survey:

The correct answer is *third selection*.

•

A satellite is currently in a circular orbit with radius 32,714 km about the center of the earth. (Note: The circumference of a circle is given by $C = 2\pi r$, where r is the radius of the circle and, to the nearest 0.01, $\pi \approx 3.14$.)

Select for 0.5 *km* the increase, to the nearest whole kilometer, in the distance the satellite travels about the center of the earth during each revolution if the orbital radius of the satellite is increased by 0.5 km. And select for 1.5 *km* the increase, to the nearest whole kilometer, in the distance the satellite travels about the center of the earth during each revolution if the orbital radius is increased by 1.5 km. Make only two selections, one in each column.

0.5 km	**1.5 km**	
		3
		9
		27
		81
		150
		450

Answer Explanation:

Infer

(Note: The circumference of a circle is given by $C = 2\pi r$, where r is the radius of the circle and, to the nearest 0.01, $\pi \approx 3.14$.)

0.5 km:

The increase in distance the satellite travels is the increase in the circumference of a circle when the circle's radius is increased from 32,714 km to

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